

Class Notes: Phylum Arthropoda

Animal Kingdom • PU-I Biology • Class Notes

Body Plan and Organisation

largest phylum in Animalia, comprising over two-thirds of known animal species, mostly insects; bilaterally symmetrical, triploblastic, coelomate (haemocoel — coelom filled with blood). Body covered by chitinous exoskeleton with distinct head, thorax and abdomen; have jointed appendages. Respiration through gills/book gills (aquatic) or book lungs/tracheal system (terrestrial); circulatory system is open; excretory organs are Malpighian tubules.

Respiration, Excretion and Examples

Sensory organs include compound and simple eyes, antennae and statocysts (balance organs). Mostly dioecious with internal fertilisation and direct or indirect development. Examples: Apis (honey bee), Bombyx (silkworm), Laccifer (lac insect); vectors like Aedes, Anopheles, Culex; pests like Locusta and cockroach; Limulus (king crab) is a living fossil.